231501053

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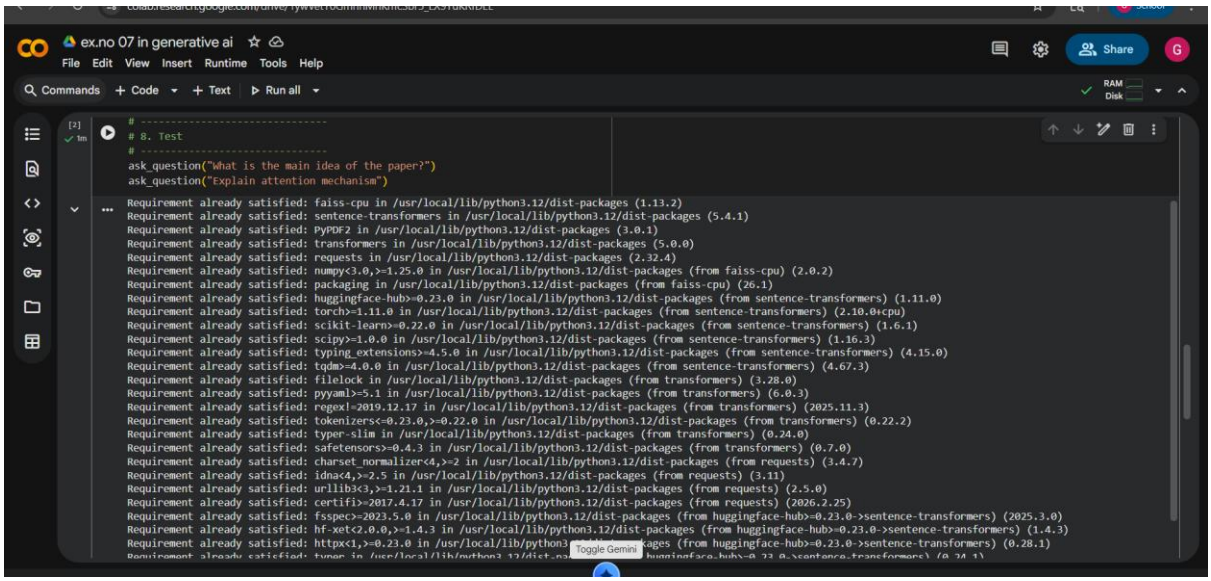
Generative AI

ex.no:07



ex.no 07 in generative ai

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```
    reader = PdfReader(file_path)
    text = ""
    for page in reader.pages:
        if page.extract_text():
            text += page.extract_text() + "\n"
    return text

document_text = load_pdf(pdf_path)

# ------------------------------
# 3. Chunk Text
# ------------------------------
def chunk_text(text, chunk_size=300):
    words = text.split()
    return [" ".join(words[i:i+chunk_size]) for i in range(0, len(words), chunk_size)]

chunks = chunk_text(document_text)

# ------------------------------
# 4. Embeddings
# ------------------------------
model = SentenceTransformer('all-MiniLM-L6-v2')
embeddings = model.encode(chunks)
embeddings = np.array(embeddings).astype("float32")

# ------------------------------
# 5. FAISS Index
# ------------------------------
index = faiss.IndexFlatL2(embeddings.shape[1])
index.add(embeddings)
```

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```
# Install libraries
!pip install faiss-cpu sentence-transformers PyPDF2 transformers requests

import faiss
import numpy as np
import requests
from PyPDF2 import PdfReader
from sentence_transformers import SentenceTransformer
from transformers import pipeline

# ------------------------------
# 1. Download PDF from URL
# ------------------------------
url = "https://arxiv.org/pdf/1706.03762.pdf"  # You can change this URL
pdf_path = "/content/sample.pdf"

response = requests.get(url)
with open(pdf_path, "wb") as f:
    f.write(response.content)

print("PDF downloaded successfully!")

# ------------------------------
# 2. Load PDF
# ------------------------------
def load_pdf(file_path):
    reader = PdfReader(file_path)
    text = ""
```

Toggle Gemini

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```
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length`(=200) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (ht
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length`(=200) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (ht

--- ANSWER ---

    Answer based on the context:

    and Beatrice Santorini. Building a large annotated corpus of english: The penn treebank. Computational linguistics , 19(2):313–330, 1993. [26] David McClosky, Eugene Cha
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is what we are missing , in my opinion . <EOS> <pad> The Law will never be perfect , but its application should be just - this is what we are missing , in my opinion . <EOS>

    Question: What is the main idea of the paper?
    Answer:
     2016. doi:10.1537/a005085 [29] Paulus, Caiming, and S. P. Bruecken. A neural network for semantic processing. arXiv preprint arXiv:1705.04305, 2016. doi:10.1537/a005085

--- ANSWER ---

    Answer based on the context:

    3.2.3 Applications of Attention in our Model The Transformer uses multi-head attention in three different ways: •In "encoder-decoder attention" layers, the queries come
positions, since for any fixed offset k,PEpos+kcan be represented as a linear function of PEpos. We also experimented with using learned positional embeddings [ 9] instead,
is what we are missing , in my opinion . <EOS> <pad> The Law will never be perfect , but its application should be just - this is what we are missing , in my opinion . <EOS>

    Question: Explain attention mechanism
    Answer:
     layers are used in the main model. The encoder is a special case of a "decoder-decoder" that is a "decoder-decode" that is the result of a "decoder-decode" in the main
```